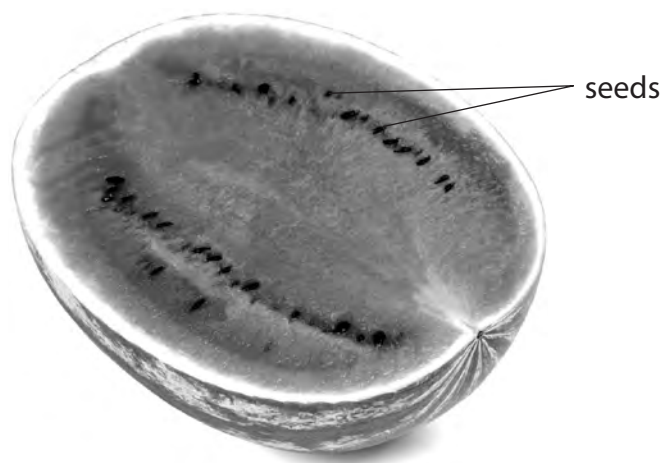


- 4 The photograph shows half a watermelon fruit, *Citrullus lanatus*, containing seeds.



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Watermelon plants are grown in all parts of the world.

Watermelon seeds can be diploid with 22 chromosomes or tetraploid with 44 chromosomes.

Tetraploid seeds have different food stores and thicker seed coats than diploid seeds.

Watermelon seeds respire their food stores during germination to develop into seedlings.

A student formed the following hypothesis:

Tetraploid seeds have a higher respiratory quotient (RQ) than diploid seeds

Plan an investigation to find evidence to support or reject this hypothesis.

(a) Describe preliminary practical work that you might undertake to ensure your proposed method would provide quantitative results.

(2)

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.

(b) Devise a detailed method, including how you would control and monitor important variables.

(9)

This image shows a full page of white paper with horizontal dotted lines. The lines are evenly spaced and run across the width of the page, providing a guide for handwriting practice. There are no margins, text, or other markings on the page.



(c) Describe how your results should be recorded, presented and analysed in order to draw conclusions from your investigation.

(3)

[illegible]

(d) Suggest **two** limitations of your proposed method.

(2)

(Total for Question 4 = 16 marks)

TOTAL FOR PAPER = 50 MARKS